

PROJECT SPECIFICATION

AI Environmental Impact Tracker

Working app name: Sprout or I/mpact

A friendly, AI-powered platform for measuring, understanding, planning, and reducing personal environmental impact.

Core promise

Make environmental impact understandable and actionable. Ask only for information that meaningfully improves accuracy, use measured data whenever possible, and help users turn their results into realistic weekly action.

Primary metrics: net water consumption, net electricity consumption, and net carbon footprint (CO2e).

1. Overview

The goal of this project is to build an AI-powered environmental impact tracker that enables anyone to calculate, understand, compare, plan, and reduce their personal environmental footprint.

The application should be free, easy to use, optimistic, and designed for the general public, with a particular focus on South African and broader African households. A user with no environmental knowledge should be able to complete an assessment in less than 10 minutes while still receiving a realistic estimate of their impact.

The application should reflect local realities such as municipal utilities, prepaid meters, loadshedding, backup generators, solar inverters, mixed household energy sources, and regional differences across South Africa and Africa.

The application measures three primary environmental metrics:

- Net water consumption
- Net electricity consumption
- Net carbon footprint (CO₂e)

The objective is not only to calculate a footprint. The application should educate users, show where their impact comes from, make the results relatable, track progress over time, create a sense of community, and provide practical AI-guided support for improvement.

2. Core Principles

2.1 Simplicity and meaningful inputs

Users should answer as few questions as possible. Every input must pass a relevance test:

- Ask the question only if the answer significantly improves the accuracy of the calculation, changes the recommendation, or is required for a core feature.
- Do not ask for information that can reasonably be inferred, calculated, looked up from an approved dataset, or extracted from an uploaded bill.
- Use progressive disclosure: estimation questions should only appear when measured data is unavailable.
- Avoid duplicated inputs across water, electricity, and carbon sections when one answer can be reused.

2.2 Accuracy

Whenever possible, calculations should be based on measured data instead of estimates.

- Utility bills, meter statements, prepaid records, and direct consumption values are preferred over behavioural estimates.
- Vehicle databases are preferred over asking users to know fuel efficiency.
- Airport databases should be used to calculate flight distance instead of asking users to estimate it.

- Regional electricity, water, and emissions datasets should be used where trustworthy local data is available.
- If data is missing, use trustworthy sources such as government agencies, utilities, universities, research institutions, and established scientific datasets.

2.3 Accessibility

The application should be suitable for people with little environmental knowledge. Questions and explanations should use simple language, and unfamiliar concepts should include short, optional explanations.

2.4 Transparency

Users should understand how their footprint was calculated, which inputs were measured or estimated, and which activities contribute most to their impact. The application should not present uncertain estimates as exact facts.

2.5 Optimism and agency

The tone should be friendly, practical, and encouraging without hiding poor performance. The product should make clear that individual action is not the only solution to environmental problems, but that personal choices can still reduce resource use, save money, influence households and communities, and contribute to broader changes in demand and norms.

3. Product Identity and Design Language

The final product name is undecided. The two current options are:

- **Sprout** - friendly, organic, simple, and growth-oriented.
- **I/npact** - modern and reflective, emphasizing both personal impact and the possibility of change.

The interface should use a consistent visual language:

- Primary palette: light blue, green, white, and soft grey.
- Friendly rounded cards, clear spacing, approachable illustrations, and simple iconography.
- A fun and lightly playful design without making the scientific results feel childish.
- Optimistic microcopy and progress feedback. Avoid guilt-heavy or fear-based messaging.
- Nature-inspired motion and visuals that react to the user's current impact score.

4. Overall User Journey

1. Open the app and understand its purpose from the home page.
2. Create an account or sign in.
3. Choose account privacy: public or private.
4. Choose an assessment method: guided questionnaire or AI conversation.
5. Provide measured environmental data, upload bills, or answer fallback estimation questions.
6. Calculate water, electricity, and carbon metrics using a deterministic calculation engine.
7. View the feedback dashboard with KPIs, graphs, comparisons, explanations, and weekly goals.

8. Use the AI planning assistant to discuss practical ways to improve.
9. Save results to the user history and update usage patterns weekly or monthly.
10. Build streaks, complete goals, unlock achievements, and optionally participate in public profiles, stories, and rankings.

5. Accounts, Privacy, and Profiles

5.1 Authentication

Users should be able to create an account and sign in securely. Environmental assessments, goals, historical results, achievements, and streaks should be linked to the user account.

5.2 Public or private account

Users should be able to choose whether their account is public or private and should be able to change this preference later.

- Private account: personal results and profile statistics are visible only to the user.
- Public account: approved profile statistics can appear on the profile page, Explore page, leaderboard, and real-life stories page.
- Public status must be based on explicit consent. Sensitive raw data, uploaded bills, exact household details, and private conversations must never be exposed publicly.
- Users should be able to withdraw public consent, after which their public profile and related social appearances should be hidden.

5.3 Profile page

Each user should have a profile page. It should display:

- Display name and optional profile image.
- How long the user has been on the app.
- Current impact score or rating.
- Water, electricity, and carbon performance summaries.
- Current weekly or monthly streak.
- Achievements and badges earned.
- Goals completed and recent progress.
- For public users only: the subset of statistics approved for public display.

6. Two Assessment Methods

The application should allow users to complete the assessment in one of two ways. Both methods must collect the same underlying data and produce identical calculations and final results.

6.1 Method 1 - Guided Questionnaire

The application asks structured questions using forms, sliders, dropdowns, checkboxes, and file uploads. This method is intended for users who prefer a traditional questionnaire.

6.2 Method 2 - AI Conversation

The user chats naturally with an AI assistant. The assistant should ask only relevant follow-up questions until enough information has been collected. It should extract structured values from natural-language answers and store them in the same data schema used by the guided questionnaire.

Important requirement

The conversational assistant must not ask every possible question. It should use the same conditional logic as the questionnaire and stop once the minimum meaningful data has been collected.

7. General Information

Collect only the general information required to choose appropriate regional factors and normalize household data:

- Household size.
- Country.
- Province, state, or region - optional unless needed for a regional factor.
- Municipality or local utility - optional unless needed for bill interpretation or local factors.

Household-level water and electricity values should be divided or otherwise normalized appropriately when presenting per-person impact. The app should clearly distinguish household totals from personal estimates.

8. Water Assessment

8.1 Preferred option - bill or statement upload

Users should be able to upload an image or digital copy of a municipal water bill or water statement. The image-processing system should extract relevant fields, including:

- Billing period or statement dates.
- Measured water consumption, preferably in litres or kilolitres.
- Water bill amount, where available.
- Meter readings, where useful for validation.
- Municipality or utility name, where it can be identified reliably.

The user must be shown the extracted values and be allowed to correct them before calculations are finalized. Actual measured consumption should always take priority over bill-cost conversion or behavioural estimates.

8.2 Manual entry

If no bill is uploaded, allow the user to enter monthly water consumption directly. A monthly water bill amount may be accepted as a fallback only when a defensible tariff or conversion method is available.

8.3 Rainwater harvesting

Ask whether the household uses rainwater harvesting. If yes, ask for the approximate percentage of household water supplied by rainwater. Rainwater should be treated as having a significantly lower environmental burden than municipally supplied water, using a transparent assumption or factor.

8.4 Estimation questions - only when needed

Only when reliable consumption data is unavailable, ask a minimal set of questions such as:

- Average shower duration.
- Number of showers per week.
- Garden irrigation.
- Swimming pool ownership.

Additional behavioural questions should only be added if testing shows that they materially improve the estimate.

9. Electricity Assessment

9.1 Preferred option - bill, receipt, or meter record upload

Users should be able to upload images of electricity bills, prepaid electricity receipts, or meter statements. The AI extraction system should identify relevant values such as:

- Billing or purchase period.
- Electricity consumption in kWh, when shown.
- Electricity bill or prepaid purchase amount.
- Tariff, meter, municipality, or utility information when reliably identifiable.
- Multiple prepaid purchases within a period, where the user uploads more than one receipt.

The app should verify and normalize units, show extracted values for user confirmation, and prioritize measured kWh over monetary conversion.

9.2 Manual entry

If no bill or receipt is available, allow users to enter monthly electricity consumption directly. Monthly bill amount may be used only as a fallback where a suitable tariff method is available.

9.3 Renewable electricity

Ask whether the household generates renewable electricity. Options include Solar, Wind, and None. If renewable electricity is used, estimate the percentage of household electricity supplied by renewable sources.

9.4 Backup power and loadshedding

Ask whether the household uses backup power during outages. Options include Generator, Inverter and battery, Solar battery backup, UPS, and None.

- For generators: collect fuel type and either average hours used per month or litres of fuel used per month.
- For inverter, battery, or solar backup: collect the approximate share of electricity supplied during outages, whether it powers the whole home or essential loads only, and whether charging comes from the grid, solar, or both.

Backup power should be reflected in electricity-related analysis and, where appropriate, in the carbon footprint.

9.5 Estimation questions - only when measured data is unavailable

- Home size.
- Electric geyser.
- Air conditioner.
- Electric stove.
- Pool pump.

10. Carbon Footprint Assessment

The carbon footprint should include transport, household energy, flights, heating, diet, food waste, shopping or consumption habits, backup power, and offsets where data is available.

10.1 Flights

Allow users to add multiple flights. For each flight collect:

- Departure airport.
- Arrival airport.
- Cabin class.
- One-way or return.
- Number of trips per year.

Flight distance should be calculated automatically from airport coordinates or an approved flight-distance data source.

10.2 Personal vehicle

- Vehicle manufacturer.
- Vehicle model.
- Model year.
- Annual distance driven.
- Average number of passengers.

The application should determine fuel type and fuel economy automatically whenever a reliable vehicle database is available. Manual fuel economy should be a fallback.

10.3 Public transport

Collect transport type and average distance travelled each week. Supported transport types should include Bus, Train, Minibus taxi / taxi, Subway, and Other.

10.4 Home heating

Collect primary heating method, months used annually, and average hours used per day. Heating methods may include Electricity, Gas, Coal, Wood, Paraffin / kerosene, Heat pump, and None.

10.5 Diet

Allow users to choose the description that best matches their diet: Heavy meat eater, Average diet, Mostly chicken, Pescatarian, Vegetarian, or Vegan.

10.6 Food waste

Estimate the share of purchased food that is wasted. Suggested choices: None, Very little, Around 10%, Around 20%, or More than 30%.

10.7 Shopping habits

Ask users to compare their purchasing habits with the average person: Much less, Less, Average, More, or Much more. This represents clothing, electronics, household goods, and similar consumption.

10.8 Carbon offsets

Ask whether users purchase certified carbon offsets. If yes, collect the approximate annual amount of carbon offset, preferably in tonnes of CO₂e or another clearly convertible unit.

11. Artificial Intelligence and API Requirements

Artificial intelligence serves four distinct roles. These roles should be separated from the deterministic calculation engine.

11.1 Conversational data collection

Use an LLM through an API. The API should receive the current structured assessment state plus the conversation and return structured data for any newly collected values, missing fields that still matter, and the next concise follow-up question when needed.

- Use structured JSON output or function/tool calling where supported.
- Validate extracted values against the application schema before saving them.
- Do not allow the LLM to invent consumption values when the user has not supplied enough information.
- Keep API keys server-side or in environment variables/secrets. Never expose keys in client code or the repository.
- Provide clear fallback behaviour if the API is unavailable.

11.2 Environmental analysis

After deterministic calculations are complete, send the calculated results and relevant user context to the LLM API. The AI should:

- Explain the user's environmental footprint in simple language.

- Identify the largest contributors.
- Explain why those contributors matter.
- Recommend practical improvements.
- Prioritize recommendations by likely impact, feasibility, and cost or effort.
- Estimate the environmental benefit of each recommendation where the calculation engine can support a defensible estimate.
- Avoid making unsupported claims or presenting uncertain estimates as exact.

11.3 AI planning assistant

The application should include a dedicated planning chat where users can discuss how to improve their impact. This feature should use an LLM API and should be grounded in the user's current results, historical progress, goals, and stated constraints.

- The user can ask questions such as how to reduce water use without changing their garden, how to lower electricity use on a budget, or how to plan lower-carbon transport for the week.
- The assistant should translate broad intentions into specific, measurable, realistic actions.
- The assistant should help revise weekly goals when circumstances change.
- The assistant should not directly overwrite user data or goals without explicit confirmation.

11.4 AI image extraction for bills and receipts

Use a multimodal AI model or document-extraction API capable of interpreting uploaded images. It should extract relevant values from water bills, electricity bills, prepaid receipts, and statements.

- Return extracted fields in a strict structured schema.
- Preserve the original image privately and do not expose it on public profiles.
- Show confidence or flag uncertain fields when the provider supports it.
- Require user review and correction of extracted values before final submission.
- Use traditional parsing or OCR rules where more reliable for a known bill format; the AI model should not be the only possible extraction path.

Architecture rule

The LLM is an advisor and interpreter, not the source of truth for arithmetic. Environmental calculations must use established conversion factors and deterministic code so the same inputs produce the same outputs.

12. Information and Education Page

The application should include a dedicated information page that educates users without requiring them to read long technical documents. It should be visually engaging and divided into three clear topics.

12.1 Greenhouse gases

- Explain what greenhouse gases are and what CO₂e means.
- Explain the main personal sources covered by the app, including energy, transport, flights, diet, and consumption.

- Explain why cumulative emissions matter and why reducing high-impact activities can make a measurable difference.

12.2 Electricity

- Explain the difference between electricity consumption and the emissions associated with generating that electricity.
- Explain why grid mix, renewable energy, efficiency, peak use, backup generation, and household equipment matter.
- Connect lower electricity use to practical outcomes such as cost savings and reduced demand.

12.3 Water

- Explain direct household water use and the energy and infrastructure required to collect, treat, pump, heat, and dispose of water.
- Explain water scarcity and why local context matters.
- Show practical examples of high-impact and low-impact changes.

12.4 Why users can make a difference

The page should explain that environmental problems require action from governments, companies, cities, and infrastructure systems as well as individuals. The user should still understand that personal changes can reduce direct resource use, save money, influence household behaviour, create social norms, and contribute to demand for lower-impact products and systems.

13. Feedback Dashboard and Progress Tracking

After completing an assessment, users should receive a clear feedback dashboard rather than a single score.

13.1 Core KPI cards

The dashboard should show separate KPI-style cards for:

- Water consumption.
- Electricity consumption.
- Carbon footprint.

Each card should show the current value, comparison with the previous period or previous assessment, and comparison with an appropriate benchmark where available.

13.2 Historical comparison

Users should be able to compare their current impact with previous statistics. The dashboard should show change over time and make direction clear, for example percentage improvement or deterioration since the previous assessment and since the user joined the app.

13.3 Separate graphs for each metric

Provide separate trend graphs for water, electricity, and carbon. Do not combine unrelated units on one axis. Graphs should support weekly, monthly, and longer-term views when enough historical data exists.

13.4 Unit period dropdown

Users should be able to choose whether statistics are displayed per day, week, month, or year. The selected period should consistently update cards, graphs, breakdowns, and comparisons where mathematically appropriate.

13.5 Relevant and relatable comparisons

In addition to exact values, the dashboard should generate relatable comparisons. Examples include:

- Your annual water use is roughly equivalent to the volume of a small swimming pool.
- Your electricity use this month could power a typical efficient refrigerator for approximately a certain period.
- Your annual carbon footprint is comparable to a certain number of return trips on a relevant route, where the comparison is based on a defensible factor.

These comparisons must be based on defined conversion factors and should be selected for relevance and scale. Avoid absurd comparisons that are technically true but not useful.

13.6 Benchmarks and context

Where reliable data is available, compare results against South African, African, country-level, and global averages. The app should clearly label the benchmark source and geographic scope. A benchmark should not be shown if the underlying comparison is misleading or unsupported.

13.7 Specific weekly goals

Feedback should include a small number of specific weekly goals based on the largest realistic improvement opportunities. Goals should be measurable and practical, for example:

- Reduce shower time by a total of 15 minutes across the week.
- Replace two short car trips with walking, cycling, or public transport.
- Reduce geyser heating time or adjust its schedule where appropriate.
- Plan one lower-impact meal change that fits the user's existing diet rather than prescribing an unrealistic full lifestyle shift.

The AI may help phrase and personalize goals, but the expected environmental saving should come from deterministic calculations where possible.

14. Reminders and Reassessment

Users should be encouraged to keep their data current. During onboarding or settings, users should choose a reminder cadence: weekly or monthly.

- Remind users to update recent usage patterns, add new bills or receipts, and confirm whether major habits changed.
- Do not force users to repeat the full assessment when only a few inputs have changed.
- Use the previous assessment as the starting point and ask only for changed or expired information.
- Allow users to snooze or disable reminders.

15. Community, Stories, and Leaderboard

15.1 Real-life stories

Create a page that shows real-life stories of users who reduced their impact through the app. Stories may only be shown when users have consented to a public account and separately consented to having their story featured.

- Show the starting challenge, actions taken, time period, and measured improvement.
- Prefer verified in-app statistics over purely anecdotal claims.
- Allow users to share practical lessons and realistic obstacles.
- Do not expose sensitive household or billing information.

15.2 Explore and public profiles

A public Explore page should allow users to discover public profiles, progress stories, and achievements. It should focus on inspiration and practical change, not only competition.

15.3 Leaderboard

Provide a public ranking system for users who opt in. Rankings should be designed carefully so users with naturally different household circumstances are not unfairly compared.

- Possible ranking views: improvement percentage, goal completion, streak length, and impact score.
- Prefer improvement-based rankings over raw lowest-consumption rankings when this creates a fairer comparison.
- Allow filters such as country or region only when privacy and sample size are sufficient.
- Public users should control which approved summary statistics appear.

16. Mascot and Reactive Environmental Visuals

16.1 Mascot concept

The app should include a single daisy flower growing on top of a small Earth. This mascot visually reflects the user's current environmental impact relative to appropriate national averages or benchmarks.

- Better-than-average impact: the Earth is clear and bright, the sky is sunny, colours are healthy, and the daisy stands upright and grows.
- Near-average impact: the scene remains mostly healthy with mild clouds or reduced vibrancy.

- Worse-than-average impact: the Earth becomes darker and cloudier, heat or fire effects gradually appear, and the daisy begins to droop or wilt.
- The transition should be gradual rather than binary. The user should see progress as their impact improves.
- The mascot should never shame the user. Poor performance should be paired with clear next actions and a visible path to recovery.

16.2 Live or semi-live nature wallpaper

The broader app background should include a live or semi-live nature scene that reacts to the user's current impact in a similar way to the mascot.

- Good impact: clearer sky, sunlight, healthy vegetation, subtle movement, and brighter atmosphere.
- Poor impact: more haze, cloud, muted vegetation, dryness, or heat effects.
- Transitions should be lightweight enough for the target platform and should not interfere with readability or accessibility.
- The wallpaper should reinforce the app's fun design language while keeping the foreground interface clean and legible.

17. Achievements, Goals, and Streaks

17.1 Achievements

Users should unlock achievements when they complete meaningful actions or maintain good performance. Examples include:

- First assessment completed.
- First verified bill uploaded.
- Completed all weekly goals.
- Reduced one metric for three consecutive reporting periods.
- Reached a personal best impact score.
- Maintained a long update streak.
- Reduced a major footprint contributor by a meaningful percentage.

Achievements should reward actual progress and consistent engagement rather than encourage meaningless clicking or data entry.

17.2 Streak feature

Track how many consecutive weeks or months a user has kept their profile updated or completed the required check-in. The streak cadence should match the user's selected reminder cadence.

- Weekly users build a weekly streak.
- Monthly users build a monthly streak.
- Streak status should appear on the profile page and dashboard.
- A missed update should not necessarily erase all historical recognition; preserve best streak as well as current streak.

18. Calculation Philosophy

The application should calculate environmental impact using established conversion factors and scientific datasets. Calculations must be deterministic and reproducible.

- Measured data always takes priority over estimated data.
- Behavioural estimates should only be used when measured values are unavailable.
- Where possible, use South African and African regional datasets, including local electricity grid factors, municipal water data, transport assumptions, and country-specific emissions factors.
- Store factor source, geography, unit, and version or date where practical so calculations can be audited and updated.
- The app should distinguish between direct consumption metrics and environmental impact metrics. For example, kWh is not the same thing as electricity-related CO₂e.

19. Suggested Page Structure

The application should contain at least the following pages or functional areas:

Page	Purpose
Home	Purpose, value proposition, explanation of the three metrics, sign-up and assessment entry points.
Sign Up / Sign In	Authentication and account creation.
Assessment	Guided questionnaire with conditional inputs and bill uploads.
AI Assessment Chat	Conversational collection of the same structured assessment data.
Feedback Dashboard	KPIs, exact metrics, trends, benchmarks, relatable comparisons, breakdowns, and weekly goals.
AI Planner	LLM API chat grounded in user data, goals, and historical progress.
Information	Education on greenhouse gases, electricity, water, and why change matters.
Profile	Membership duration, score, streaks, achievements, goals, and public statistics where permitted.
Explore / Stories	Public profiles and consented success stories.
Leaderboard	Opt-in rankings focused on improvement, goals, streaks, and impact.
Settings	Privacy, reminder cadence, unit period defaults, account settings, and consent controls.

20. Data and State Requirements

The application should maintain structured data so that questionnaire inputs, chat inputs, image extraction, calculations, history, goals, and social features all use consistent records.

- User record: account, privacy status, join date, reminder cadence, and unit-period preference.
- Assessment record: timestamp, reporting period, source type for each input, measured-versus-estimated flag, and calculated outputs.
- History record: water, electricity, carbon, overall score, and category breakdowns over time.
- Goal record: target action, metric, target value, start date, end date, status, and completion result.
- Achievement record: achievement type, date earned, and evidence or trigger.
- Streak record: cadence, current streak, best streak, and last valid check-in.
- Public profile settings: exact fields that the user has allowed to be shown publicly.
- Story consent and story record: separate from general public-account consent.
- Uploaded bill record: private file reference, extraction result, user-corrected values, and validation status.

21. Dashboard Output Requirements

The dashboard should summarize results in a way that is useful both immediately and over time. It should include:

- Total water consumption.
- Total electricity consumption.
- Total carbon footprint.
- Category breakdowns and largest contributors.
- National and other relevant benchmark comparisons where supported.
- Change from previous period and change since joining.
- Separate historical graphs for each metric.
- Per day, week, month, or year display selector.
- Relatable equivalence comparisons.
- AI-generated interpretation.
- Specific weekly goals.
- Progress toward active goals.
- Current streak and recent achievements.
- Reactive mascot and nature visuals linked to current impact.

22. Long-Term Vision

The application should become a personal environmental companion rather than a once-off calculator. The immediate product should already support historical tracking, goals, achievements, streaks, AI planning, bill-image extraction, reminders, and optional community features. Future extensions may include:

- Community challenges.

- Household or school teams.
- Smart meter and prepaid meter integrations.
- Utility-provider integrations.
- Solar inverter and backup-power integrations.
- Personalized periodic progress reports.
- More detailed regional environmental datasets across South Africa and Africa.
- More robust verified-story workflows and community moderation.
- Privacy-preserving neighbourhood or city comparisons where sample sizes are sufficient.

The overall vision is to create an accessible, scientifically grounded, visually engaging platform that empowers individuals to understand and reduce their environmental impact through accurate measurement, education, progress tracking, social inspiration, and actionable AI-driven guidance.

Appendix A. Non-Negotiable Implementation Rules

1. Ask only relevant and meaningful questions.
2. Prefer measured values over estimates.
3. Use the same underlying schema and calculation engine for questionnaire and chat input.
4. Use an LLM API for conversational collection, analysis, and planning; use a multimodal model or document-extraction API for bill images.
5. Never let the LLM perform the authoritative footprint arithmetic.
6. Require users to review AI-extracted bill data before calculation.
7. Keep uploaded bills, raw household data, and private chats private regardless of public profile status.
8. Require explicit consent for public profiles and separate consent for featured stories.
9. Show exact metrics together with context, trends, and relatable comparisons.
10. Provide specific weekly goals and allow the user to discuss and revise plans through the AI planner.
11. Support weekly or monthly reminders and corresponding streaks.
12. Use the light blue, green, white, and grey visual theme consistently.
13. Make the mascot and environmental wallpaper react gradually to the user's impact relative to appropriate benchmarks.
14. Keep the tone friendly and optimistic while remaining honest about poor performance and uncertainty.